



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING -(24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Powered Adaptive Learning and Personalized Course Recommendation System	
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Abstract

AI-Powered Adaptive Learning and Personalized Course Recommendation System is an intelligent educational platform designed to provide personalized learning experiences based on individual student performance, interests, and learning progress. Traditional learning platforms generally provide the same courses and learning paths to all students, without sufficiently considering differences in their knowledge levels and learning needs. The proposed system addresses this limitation by continuously analyzing student performance and dynamically adapting course recommendations.

The system collects information such as quiz scores, completed courses, learning interests, and topic-wise performance to identify a student's strengths, weaknesses, and current learning level. Using Artificial Intelligence and a recommendation mechanism, it suggests suitable courses, learning resources, and prerequisite topics based on the student's profile. The difficulty of recommended content can be automatically adjusted as the student's performance improves or declines. For example, students with low performance in a particular topic may receive beginner-level courses and prerequisite materials, while students demonstrating strong performance can be recommended advanced courses.

The platform also maintains a personalized learning profile and continuously updates recommendations based on new assessment results and learning activities. **DevOps practices**, including Git-based version control, automated testing, Continuous Integration and Continuous Deployment (CI/CD), Docker-based deployment, and application monitoring, can be incorporated to support continuous development and reliable delivery. By combining **Artificial Intelligence, Adaptive Software Engineering, Personalized Learning, and DevOps**, the proposed system aims to improve student engagement, learning efficiency, and academic performance through a continuously evolving and personalized learning environment.

Signature of the Guide

HOD